

Dynamic RAM

Larger microcomputer systems use Dynamic RAM (DRAM) rather than Static RAM (SRAM) because of its lower cost per bit. DRAMs require more complex interface circuitry because of their multiplexed address bus and because of the need to refresh each memory cell periodically.

After this lecture you should be able to: (1) describe basic DRAM structure and terminology, (2) interface DRAMs to a CPU bus by multiplexing row and column address lines and forcing refresh cycles, and (3) justify the choice of DRAM or SRAM for a particular application.

DRAM Structure

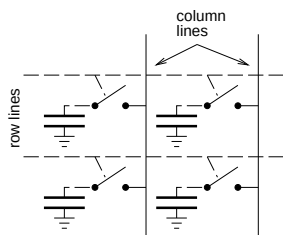
A description of the structure of a DRAM helps explain some of the unique features of DRAMs.

A typical DRAM memory is laid out as a square array of memory cells with an equal number of rows and columns. Each memory cell stores one bit. The bits are addressed by using half of the bits (the most significant half) to select a row and the other half to select a column.

Exercise: How many rows and columns would there be in the memory cell array in a 16 M by 1 (16 Mbit) DRAM?

Each DRAM memory cell is very simple – it consists of a capacitor and a MOSFET switch. A DRAM memory cell is therefore much smaller than an SRAM cell which needs at least two gates to implement a flip-flop.

When a bit in the memory array is selected, all of the capacitors in the selected row are connected to a line shared by all the memory cells in a column. The following diagram shows this using switches in place of transistors.

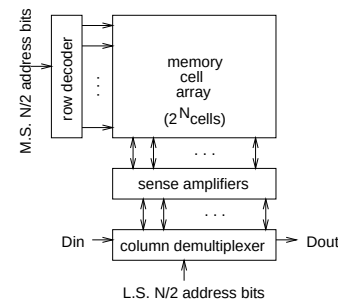


Since the charge stored in each memory cell capacitor is relatively small, each column line is connected to a “sense amplifier” which amplifies the voltage present on the line.

The row address drives a decoder which enables only one row-select line. The column address drives a multiplexer which selects one of the column lines and connects it to the input or output.

The outputs of the sense amplifiers are also fed back to the capacitors and thus re-charge them with the value stored there. Thus each row access *refreshes* the contents of that row.

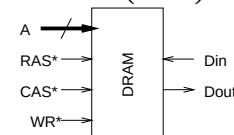
During a write, the value on the input over-rides the sense amplifier value for the addressed column and this stores new data into the desired memory cell.



Address Multiplexing

In order to reduce the number of pins on the chip and thus reduce the size of the DRAM chip, most DRAM chips multiplex the row and column addresses onto the same set of pins.

Two strobes, RAS* (row address strobe) and CAS* (column address strobe) are used to tell the chip which part of the address is currently on the address pins. To further reduce the chip count the CAS* signal typically acts as an output enable when R/W* line is high (read) and the falling edge acts as a write strobe when R/W* is low (write).



Exercise: How many address pins would be found on a typical 16 M x 1 DRAM?

DRAM Timing

In addition to the DRAM timing requirements of setup and hold times for the row and column addresses, DRAMs also require a minimum “pre-charge” time between the end of RAS* or CAS* and the start of the next cycle. This “recovery” time is needed to re-charge the storage capacitors. This pre-charge time extends the minimum cycle duration to considerably more than the access time. For example, a DRAM with a 60 ns access time may have a minimum cycle time of 100 ns.

DRAM Timing

The sequence of operations required to read or write from a DRAM both start in the same way:

- set R/W* to the appropriate value and place the row address (the MS half of the address) on the address pins,
- wait the RAS* setup time, bring RAS* low, and wait for the RAS* hold time

In order to read from the DRAM:

- place the column address on the address pins, wait the CAS* setup time, bring CAS* low, and wait for the CAS* hold time
- wait until the access times from both CAS* and RAS* are met and then read the data from the data out pin

In order to write to a DRAM the sequence is similar except that during a write cycle the data is latched on the falling edge of CAS*:

- place the column address on the address pins, data on the data input pin, wait for the CAS* and data setup times, bring CAS* low, wait for the CAS* and data hold times.

At the end of either cycle we must then bring RAS* and CAS* high and wait the pre-charge (recovery) time before starting another cycle.

Exercise: Draw a timing diagram for a DRAM read cycle showing the address lines, RAS*, CAS*, WR* and the data pins. Show on the timing diagram the following specifications: address setup and hold times from RAS* and

CAS* active, access times from RAS* and CAS* active, minimum “pre-charge” times (from RAS* or CAS* inactive), and the minimum cycle time (from RAS* to RAS*).

For a write cycle, show the setup and hold times for D_{in} from CAS* active.

Refresh

Since the DRAM storage capacitor discharges over time it must be refreshed periodically. The DRAM’s structure ensures that all the memory cells in a row are refreshed every time that row is read. Therefore it is only necessary to periodically cycle through all of the row addresses to refresh all of the bits in the memory array.

The simplest technique to provide DRAM refresh is to include a device (such as a DMA controller or video display circuit) that accesses the RAM in such a way that all of the rows are accessed at least once during the minimum refresh time (typically every few tens of milliseconds). This is called *RAS*-only refresh* because it’s not necessary to assert CAS* in order for the refresh operation to take place.

Another technique is to add a circuit that periodically forces a cycle in which CAS* is asserted before RAS*. This is called *CAS* before RAS* refresh*. Modern DRAMs have an internal refresh counter that cycles through the possible row values. On these DRAMs the CAS* before RAS* operation causes an internal row-refresh operation. The advantage of this type of refresh is that the refresh controller need only control RAS* and CAS*, it need not generate the refresh row addresses.

Exercise: Assume a microprocessor with a 200 ns memory cycle time is using a 1 MByte DRAM with a maximum refresh time of 10 ms. How many row addresses will have to be refreshed every 10ms? What is the approximate time between each refresh cycle? How many memory cycles are there per refresh cycle? What percentage of the memory accesses are “wasted” on refresh cycles?

DRAM versus SRAM

Since the SRAM devices require more circuitry per memory element they are more expensive (per bit) to produce and have lower density per chip. The typical

ratio between DRAM and SRAM for the same size chip is about 4 to 1.

The disadvantages of DRAMs are that they require additional control circuits to multiplex address lines and to handle refresh. If DRAMs are organized as bit-wide devices it is necessary to use a number of devices that is a multiple of the data bus width (8, 16 or 32) in a system.

The use of large DRAM arrays in which the CPU address and data buses must drive many chips usually requires buffers for the address and data lines. Because of these reasons DRAMs are mainly used in systems that require large memories and SRAMs are mainly used in smaller systems.

The fastest RAM designs are static, so SRAMs are often used for high-speed memories such as cache or address translation tables.

CMOS SRAMs consume very little power when not being accessed so they are often used in low-power designs. With the use of battery (or a large capacitor) backup they can retain their contents for months. On the other hand, since DRAMs must be continuously refreshed, their power consumption cannot be reduced to very low values.

Due to the larger number of bits per chip and wider bus sizes, DRAMs are now being offered in nybble (4-bit) and larger organizations. The number of extra pins required to provide the additional data bits is less of a concern with modern high density packages such as QFP and SOIC.

Exercise: Consider a system using 16 Mbit X1 memories to design a memory array for a microprocessor system with a 32-bit data bus. What is the minimum amount of memory that could be provided using these devices?

the next cycle in order to help satisfy CPU hold times.

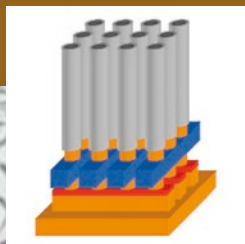
DRAM Control Circuits

A fair bit of glue logic is required to interface the CPU to a DRAM memory array. This logic needs to multiplex the CPU address lines onto the DRAM address pins and take care of refresh cycles. The control circuitry also needs to prevent the CPU from accessing the DRAM during refresh cycles. In the next assignment you will design a simple DRAM controller.

Page Mode and Other Features

Since the contents of each row are read at the end of each RAS*, it should be possible to read more than one column in this row without having to re-read the row. Some DRAMs support such a “page mode” in which multiple CAS* cycles may be used to access multiple addresses in one row (“page”) after one RAS* cycle.

Some high-speed DRAMS include an extended data out (EDO) feature, in which the output data is held past the end of a read cycle and into the start of



Future of dynamic random-access memory as main memory

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Dynamic random-access memory (DRAM) is the main memory in most current computers. The excellent scalability of DRAM has significantly contributed to the development of modern computers. However, DRAM technology now faces critical challenges associated with further scaling toward the ~10-nm technology node. This scaling will likely end soon because of the inherent limitations of charge-based memory. Much effort has been dedicated to delaying this. Novel cell architectures have been designed to reduce the cell area, and new materials and process technologies have been extensively investigated, especially for dielectrics and electrodes related to charge storage. In this article, the current issues, recent progress in and the future of DRAM materials, and fabrication technologies are discussed.

Introduction

Dynamic random-access memory (DRAM) has served as the main memory in modern computers since it was introduced by Intel Corporation in 1972. DRAM is widely used in modern computers owing to characteristics such as high-speed operation, large integration density, and excellent reliability.

A DRAM cell has a simple structure comprising one capacitor (1C) connected by one transistor (1T) (1T-1C) to the bit line (the line through which information is written to/read from the memory cell) (**Figure 1a**). The access transistor is connected to the word line and acts as a switch. The capacitor stores each bit of data as a negative or positive electrical charge. The memory state is read by sensing the stored charge on the capacitor via the bit line, which is set to $V_{cc}/2$, (V_{cc} : operating voltage of the chip) with the transistor closed. When the access transistor is on, the stored charge carriers flow into the bit line, which changes its potential. This voltage change is detected and amplified by the sense amplifier connected to the bit line.

During the past several decades, exponential growth in the number of memory cells per chip has occurred. Perpetual memory cell scaling has been the major strategy for realizing rapid increases in memory density.¹ DRAM is amenable to scaling because of its simple structure, and scaling of DRAM down to a ~100-nm technology node (specific manufacturing process and its design rule) is easily achieved with traditional

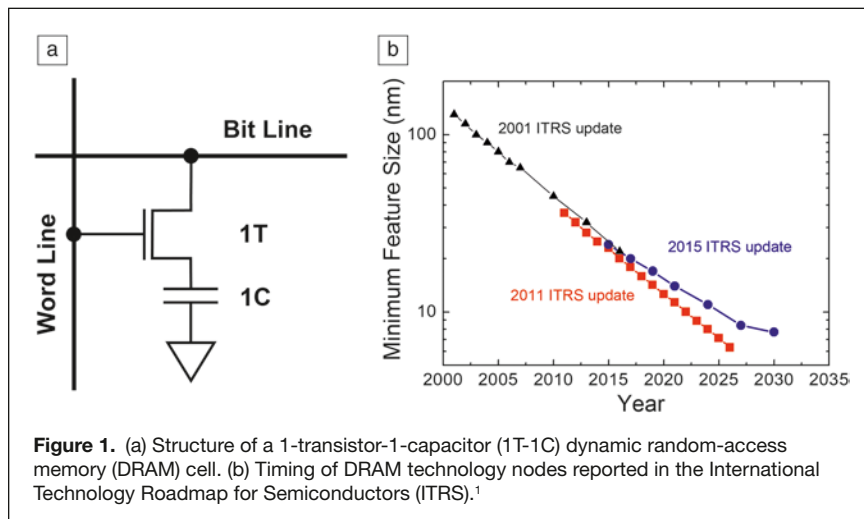
materials such as Si-based dielectrics (SiO_2 and SiN_x) and electrodes (poly-Si). This excellent scalability of DRAM has led to its long-lasting success. Today, much effort is devoted to accelerating the scaling of DRAM cells further to the ~10-nm technology node² (**Figure 1b**).

DRAM is a charge-based RAM that requires a certain cell capacitance value (~10 fF/cell) to secure the minimum voltage difference detectable by the sense amplifier, regardless of cell size. The ever-shrinking dimension of the capacitor in DRAM cells will eventually result in failure to meet this cell capacitance requirement. Hence, advancing capacitor technology is essential for continued scaling of DRAM. The scaling of the DRAM cells also raises problems in the access transistor. Without major innovations in structure, materials, and processing, scaling of DRAM will end soon.

Transistor technology in DRAM

Unlike the transistor in performance-oriented logic devices, the access transistor in DRAM requires a high ON/OFF current ratio (~ 10^8) to prevent substantial loss of the charges stored in the capacitor and to write the data within a short time (less than a few tens of a nanosecond). As the DRAM cell shrinks, obtaining sufficient data retention time becomes more challenging. The increase in the channel doping concentration with scaling of the feature size results in increased electrical field and junction leakage current. A simple and effective way

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to overcome this detrimental short-channel effect is to increase the channel length. Three-dimensional (3D) structured transistors such as recess-channel-array transistors (RCATs) have been used to increase the effective channel length in this restricted dimension (**Figure 2a**).³ Further increase in the channel length has been achieved by using modified RCAT structures such as sphere-shaped RCAT⁴ and U-shaped RCAT⁵ (**Figure 2a**). Below a technology node of ~50 nm, saddle-fin transistors, which combine fin-shaped field-effect transistor with a RCAT, have been utilized to obtain superior current-driving capability and sufficient data retention time.^{6,7}

An 8 F² (where F is the minimum feature size) cell design architecture with a folded bit-line structure⁸ has been traditionally used in DRAM because of reliable operations for the cell architecture. However, the demand for further scaling of

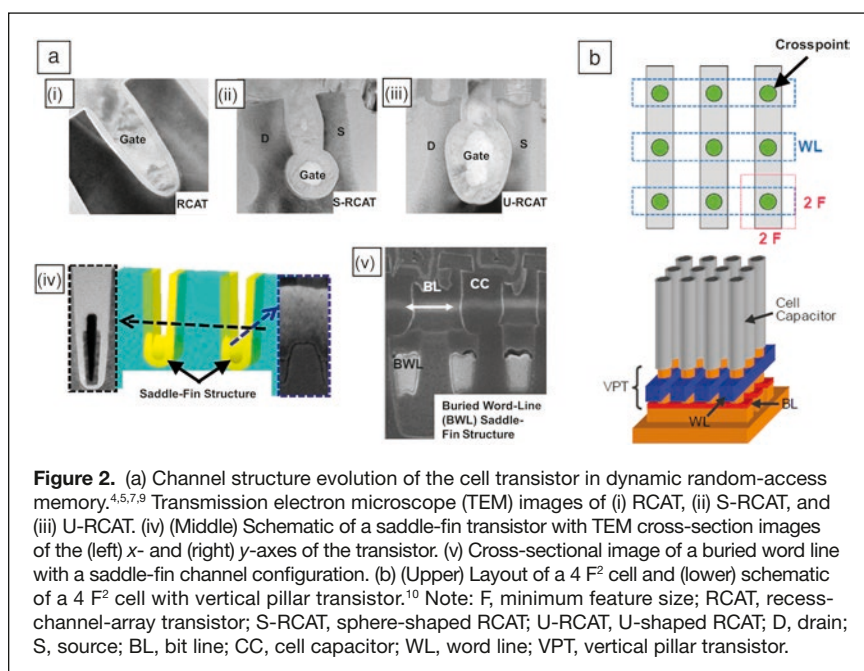
vertical pillar transistor is placed on the buried bit line and beneath the storage node (**Figure 2b**). This cell design is highly favorable to lower the bit-line capacitance, due to the distinct arrangement of the buried bit line, aside from the capacitor. However, transiting from 6 F² to 4 F² DRAM cell design is still challenging. For further scaling of DRAM cells, severe technological challenges such as the floating body effect of the channel in the vertical transistor (the channel in the vertical transistor is isolated from the Si substrate), structural vulnerability of the pillar channel, and high resistance of the bit line remain.

Capacitor technology in DRAM

For successful DRAM cell operation, the capacitor in the DRAM cell should meet two requirements—sufficient capacitance (~10 fF/cell) and ultralow leakage current ($J_g < 10^{-7}$ A/cm² at the operating voltage) irrespective of the cell size. The cell capacitance is expressed by:

$$C = \epsilon_0 k \frac{A}{t_{\text{phys}}}, \quad (1)$$

where C , ϵ_0 , k , A , and t_{phys} are the capacitance, vacuum permittivity, dielectric constant, effective capacitor area, and the physical thickness of the dielectric layer, respectively. Scaling of the DRAM cell has continuously reduced the area allocated to the capacitor in the cell, such that a 3D structured capacitor is used to obtain the necessary capacitance in the limited area, as shown in **Figure 3a**.¹¹ The aspect ratio of the capacitor has sharply increased and will reach ~100 shortly because of the aggressive scaling of DRAM. However, further increase in the aspect ratio is impossible because of the structural vulnerability of the storage node.



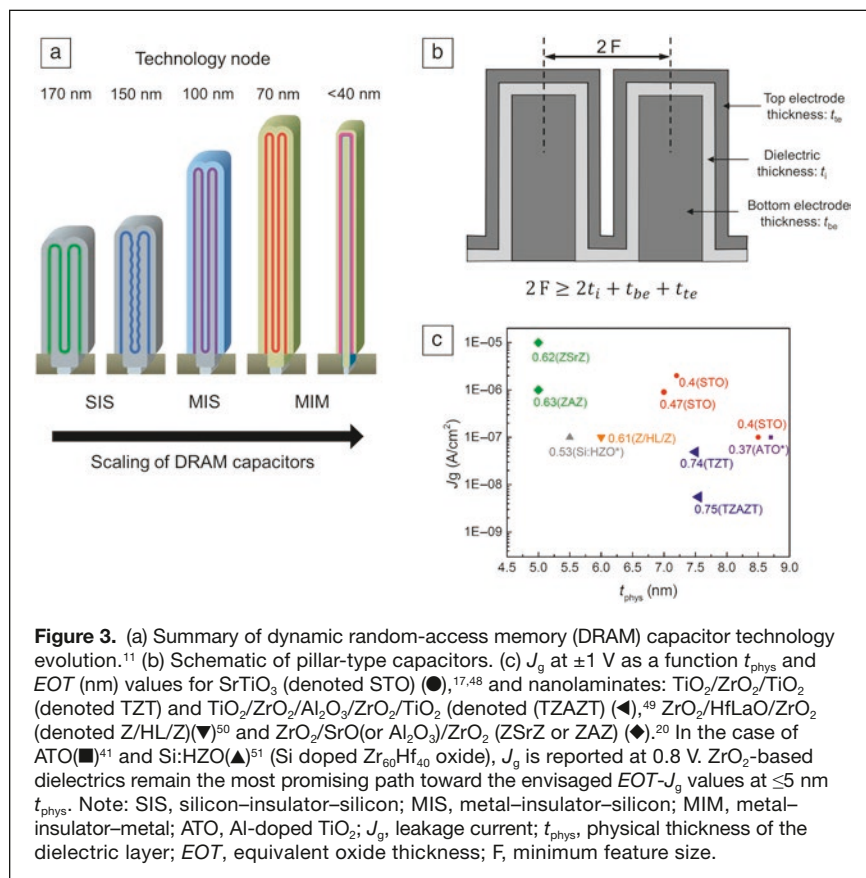


Figure 3. (a) Summary of dynamic random-access memory (DRAM) capacitor technology evolution.¹¹ (b) Schematic of pillar-type capacitors. (c) J_g at ± 1 V as a function t_{phys} and EOT (nm) values for SrTiO₃ (denoted STO) (●),^{17,48} and nanolaminates: TiO₂/ZrO₂/TiO₂ (denoted TZT) and TiO₂/ZrO₂/Al₂O₃/ZrO₂/TiO₂ (denoted TZAAT) (◀),⁴⁹ ZrO₂/HfLaO/ZrO₂ (denoted Z/Hf/Z) (▼)⁵⁰ and ZrO₂/SrO(or Al₂O₃)/ZrO₂ (ZSrZ or ZAZ) (◆).²⁰ In the case of ATO (■)⁴¹ and Si:HfO₂ (▲)⁵¹ (Si doped Zr₆₀Hf₄₀ oxide), J_g is reported at 0.8 V. ZrO₂-based dielectrics remain the most promising path toward the envisaged EOT - J_g values at ≤ 5 nm t_{phys} . Note: SIS, silicon-insulator-silicon; MIS, metal-insulator-silicon; MIM, metal-insulator-metal; ATO, Al-doped TiO₂; J_g , leakage current; t_{phys} , physical thickness of the dielectric layer; EOT , equivalent oxide thickness; F , minimum feature size.

Thinning of the dielectric layer is also not an eventual solution to acquire large capacitance as the electric field applied to the dielectric increases with decreasing t_{phys} , which leads to a significant rise in the J_g .

Therefore, a higher- k material must be used as the dielectric in the DRAM capacitor. The t_{phys} value can be increased to some extent owing to the high- k value, reducing the effective electric field at the operating voltage. The figure of merit used by DRAM capacitor technology is the electrical performance of the dielectric material as compared to SiO₂, known as the equivalent oxide thickness (EOT). The EOT of the dielectric is given by:

$$EOT = \frac{3.9 \times t_{phys}}{k}, \quad (2)$$

where 3.9 is the dielectric constant of SiO₂. In the last decade, extensive investigations have focused on identifying high- k dielectrics in conjunction with specific metal electrodes to reduce the EOT value.^{11–20}

DRAM capacitor technology is more seriously challenged than ever. Sub-20-nm DRAM technology nodes require a low EOT value of <0.5 nm, which is difficult to achieve. A significant challenge is the limit in t_{phys} of the dielectric and electrode layers. For >40 -nm technology nodes, the lateral size of the capacitor is incomparably larger than the t_{phys} of the layers. This large technology node enables the use of cylindrical

capacitors to maximize the effective capacitor area. For sub-20-nm technology nodes, however, the cylindrical structure is no longer valid and pillar capacitors are utilized because of their small feature size ($F < 20$ – 40 nm) (Figure 3a). The t_{phys} of the dielectric is stringently limited below ~ 5 nm, even in the pillar structure, because the sum of $2 \times t_{phys}$, width of the bottom electrode, and the thickness of the top electrode should not exceed $2 \times F$ (Figure 3b). A new high- k material for future DRAM capacitors should achieve both ultralow J_g (10^{-7} A/cm² at the operating voltage) and a low EOT of <0.5 nm at a low thickness of <5 nm.

Dielectrics in DRAM capacitors

The dielectric layer should be conformally formed over the 3D capacitor structure. A better step coverage is necessary at a lower t_{phys} , because the electric field at a thinner layer can be significantly influenced by even a tiny difference in the t_{phys} . Atomic layer deposition (ALD), known to achieve exceptional step coverage, is appropriate for the growth of the dielectric layer over the 3D capacitor. In addition, ALD can grow high-quality films at a relatively low temperature. Consequently, the ALD technique has been extensively used

for studies on the dielectric in DRAM capacitors.

DRAM capacitor technology has relied on the ZrO₂ dielectric for a decade as it enabled attainment of the 25-nm technology node. The trilayer structure of tetragonal (or cubic ZrO₂ ($k \approx 40$))/amorphous Al₂O₃ ($k \approx 9$)/tetragonal (or cubic ZrO₂), called ZAZ, in combination with TiN electrodes has replaced HfO₂ for the capacitors required for the ≤ 45 -nm technology node.²¹ Nevertheless, it is generally agreed that a J_g of 10^{-7} A/cm² at ± 1 V cannot be maintained with the ZAZ nanolaminate stack for a sub-20-nm technology node with smaller surface areas that involves reducing t_{phys} of the dielectric.²

Robertson²² showed that the k of oxides tends to vary inversely with the bandgap, which limits the choice of high- k oxides to several candidates comprising TiO₂ and SrTiO₃. Their bandgap is rather small (3.2–3.3 eV); however, their k values are more than 100. Although the allowable cation nonstoichiometry range of bulk SrTiO₃ is narrow and off-stoichiometric SrTiO₃ thin films usually show decreased k value, its perovskite structure could be maintained in a reasonable range of Sr:Ti ratios around stoichiometric SrTiO₃.^{14,23} Significant efforts have been devoted to the development of SrTiO₃^{14,16,17,23–33} and TiO₂^{12,13,18,34,35} by ALD. High- k values of SrTiO₃ while preserving low J_g could be achieved by controlling the stoichiometry (Sr enrichment)^{16,17} or the grain size.^{27,30}

TiO₂ crystallizes as anatase ($k \approx 40$) or rutile ($k \approx 80$) phases. Rutile, a high-temperature phase,³⁶ is incompatible

with the thermal budget required in the DRAM fabrication process ($<600^{\circ}\text{C}$). However, when grown on RuO_2 conductive oxide, rutile TiO_2 phase with similar lattice parameters crystallizes at deposition temperatures (i.e., 250°C) owing to the template effect displayed by the substrate,^{12,13,37,38} thus reducing both the EOT and J_g .^{12,15,34} Nevertheless, the scalability of t_{phys} is limited to 10–12 nm. Doping TiO_2 with Al^{11,39–42} could further reduce J_g , leading to further reduction in t_{phys} to ~ 7 nm.^{39,41}

A doping approach could also be considered for other dielectrics (e.g., HfO_2 and ZrO_2). Dopants with lower electronegativities and larger ionic radii than those corresponding to the host oxide led to higher- k and a reduction of J_g for HfO_2 ^{43–45} and ZrO_2 .^{46,47} Higher symmetry phases are formed via doping (such as tetragonal or cubic) leading to larger k , while the lower J_g was attributed to the shift of the charge states of the oxygen vacancies into the conduction bands.⁴⁷

As previously described, continued downscaling of DRAM technology toward the sub-20-nm technology node will require a low t_{phys} of ≤ 5 nm. At such a low thickness, other characteristics of the capacitors become important, including electrode and dielectric surface roughness, lattice mismatch, and chemical compatibility of electrode/dielectric. A summary of the lowest $EOT - J_g - t_{\text{phys}}$ at ± 1 V for ALD dielectrics is given in Figure 3c.^{17,20,41,48–51} This shows that the nanolaminates' approach combining a high bandgap with a high- k dielectric leads to the lowest J_g when thinning the dielectric. However, for $t_{\text{phys}} \leq 5$ nm, the nanolaminate approach is less likely to be an option because of intermixing of the layers under the thermal budget applied to the capacitor.

Comprehensive *ab initio* calculations of the bandgap and k ⁵² (namely the generalized gradient approximation for bandgap and local density approximation for k) showed that cubic BeO can display the highest- k (~ 300) and bandgap values (>9 eV). However, the cubic rock-salt structure of BeO is a high-pressure phase; therefore, stabilization under ambient conditions is required. Efforts are being devoted to depositing a high-bandgap BeO by ALD;⁵³ however, no high- k material has yet been achieved.

Electrodes in DRAM capacitor

Further reduction in J_g can also be attained by electrode engineering. The electrode in a DRAM capacitor should have a high work function and a sharp interface between the electrode and dielectric for better dielectric performance. TiN, grown by ALD using TiCl_4 and NH_3 , currently functions as the electrode in DRAM capacitors. However, the work function of TiN is insufficient to suppress J_g at the thin dielectric thickness required in the ~ 10 -nm technology node. Therefore, much effort has also been devoted to developing new electrodes, including noble metal and conducting oxide electrodes. Among noble metals, ruthenium (Ru) is considered the most promising for DRAM capacitor electrodes. Ru is favorable for suppressing J_g because of its

relatively high work function (~ 4.8 eV), and the ease with which Ru can be dry etched is also advantageous for patterning the electrodes. The use of Ru as the electrode has shown possibilities of lowering J_g .^{54–56} Forming a continuous and smooth Ru layer on oxides at a low Ru thickness of <5 nm is difficult because of its high surface energy. Morphological issues such as blisters, which often occur on ALD Ru on oxides, must also be addressed for electrode application of Ru.⁵⁷

Conducting oxides such as RuO_2 and SrRuO_3 have also attracted attention as potential electrodes. The work function of RuO_2 and SrRuO_3 is even higher than that of Ru, which is favorable for low J_g . In particular, the structural coherency with promising high- k materials (e.g., rutile $\text{TiO}_2/\text{RuO}_2$ ^{13,58,59} and $\text{SrTiO}_3/\text{SrRuO}_3$ ^{60,61}) provides concurrent decreases in EOT and J_g . However, these conducting oxides have not yet been practically used in DRAM capacitors. Those Ru-containing oxides are easily reduced during the back-end process, because of the weak bonding between ruthenium and oxygen. Ta-doped SnO_2 has recently been suggested as a reduction-resistant oxide electrode for DRAM capacitors.⁶² Capacitors composed of rutile TiO_2/Ta -doped SnO_2 stack showed both excellent dielectric properties and thermal stability in experiments.

Summary and outlook

DRAM is the representative memory used in modern computers, but it appears to be facing serious challenges for further scaling toward the ~ 10 -nm technology node. Much effort has been dedicated to prolonging its scaling. Structural modification of cell transistors has been attempted to suppress J_g and lower the bit-line capacitance. Transition of the cell architecture to 4F^2 has also been suggested for further DRAM scaling. To satisfy the stringent requirements of capacitors in ~ 10 -nm technology nodes, new higher- k dielectrics and electrodes have been extensively investigated. Although high- k oxides such as TiO_2 and SrTiO_3 show potential for further scaling of DRAM, their large J_g has to be resolved at a low thickness of <5 nm. A new dielectric material with both large bandgap (over ~ 5 eV) and large k (>50) should be designed for the ~ 10 -nm technology node. Otherwise, downscaling of DRAM might end at approximately the 15-nm technology node.

Although the scaling of DRAM will eventually end, DRAM might maintain its status as the main memory for a long time. The market for infrastructure such as servers, storage, and networking continues to grow rapidly. Bandwidth is the major challenge of DRAM for those applications. Although die stacking with conventional wire bonding has limited data-transfer rates, the emergence of high-bandwidth memory (HBM) through-silicon-via technology shows great promise with much improved data rates and reduced power consumption. Die stacking of DRAM in the form of HBM could be the future of this long-standing main memory, following the end of DRAM scaling.